Optimization Problems Using Derivatives

Calculus Worksheet · Grade 11–12 / College Calculus

Name: _____ Date: _____

Learning Objectives

- Set up two equations (constraint and objective) from a word problem
- Use substitution to reduce the objective function to a single variable and apply differentiation
- Verify maximum or minimum values using the first derivative sign chart

Problems

1. Find two positive numbers whose sum is 60 such that their product is maximized.

$$x + y = 60, \quad P = xy$$

2. Find two positive numbers such that the sum of the first and three times the second is 120, and their product is maximized.

$$x + 3y = 120, \quad P = xy$$

3. A farmer has 200 meters of fencing and wants to enclose a rectangular garden. One side of the garden borders a barn wall (no fencing needed on that side). Find the dimensions that maximize the enclosed area.

$$2W + L = 200, \quad A = LW$$

4. Find two positive numbers such that their sum is 80 and the sum of their squares is minimized.

$$x + y = 80, \quad S = x^2 + y^2$$

5. A rectangular garden is to be enclosed using 300 meters of fencing for all four sides. Find the dimensions that maximize the area.

$$2L + 2W = 300, \quad A = LW$$

6. Find a positive number such that the sum of the number and its reciprocal is minimized.

$$f(x) = x + \frac{1}{x}, \quad x > 0$$

7. A box with a square base and no top is to be made from a sheet of metal with a total surface area of 108 square centimeters. Find the dimensions of the box that maximize its volume.

$$x^2 + 4xh = 108, \quad V = x^2 h$$

8. A Norman window consists of a rectangle topped with a semicircle. If the perimeter of the window is 10 meters, find the radius of the semicircle and the height of the rectangular portion that maximize the area.

$$2h + 2r + \pi r = 10, \quad A = 2rh + \frac{\pi r^2}{2}$$

9. A cylindrical can is to hold 500 cubic centimeters of liquid. Find the radius and height of the can that minimize the total surface area (including both circular ends).

$$\pi r^2 h = 500, \quad S = 2\pi r^2 + 2\pi rh$$

10. A company sells x units of a product per week. The weekly revenue function is R(x) = 200x - 0.5x squared and the weekly cost function is C(x) = 50x + 2000. Find the number of units that maximizes the weekly profit, and state the maximum profit.

$$P(x) = R(x) - C(x) = 150x - 0.5x^2 - 2000$$

Optimization Problems Using Derivatives — Answer Key

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Answer Key

1. Answer: x = 30, y = 30; Maximum product = 900

- Constraint: x + y = 60, so x = 60 - y
- Objective: P = xy = (60 - y)y = 60y - y²
- Derivative: P'(y) = 60 - 2y
- Set P'(y) = 0: 60 - 2y = 0 → y = 30
- Sign chart: P'(29) > 0 and P'(31) < 0, confirming a maximum at y = 30
- x = 60 - 30 = 30; Maximum product = 30 × 30 = 900

2. Answer: x = 60, y = 20; Maximum product = 1200

- Constraint: x + 3y = 120, so x = 120 - 3y
- Objective: P = xy = (120 - 3y)y = 120y - 3y²
- Derivative: P'(y) = 120 - 6y
- Set P'(y) = 0: 120 - 6y = 0 → y = 20
- Sign chart: P'(19) > 0 and P'(21) < 0, confirming maximum at y = 20
- x = 120 - 3(20) = 60; Maximum product = 60 × 20 = 1200

3. Answer: W = 50 m, L = 100 m; Maximum area = 5000 m²

- Constraint: 2W + L = 200, so L = 200 - 2W
- Objective: A = LW = (200 - 2W)W = 200W - 2W²
- Derivative: A'(W) = 200 - 4W
- Set A'(W) = 0: 200 - 4W = 0 → W = 50
- Sign chart: A'(49) > 0 and A'(51) < 0, confirming maximum at W = 50
- L = 200 - 2(50) = 100; Maximum area = 100 × 50 = 5000 m²

4. Answer: x = 40, y = 40; Minimum sum of squares = 3200

- Constraint: x + y = 80, so x = 80 - y
- Objective: S = x² + y² = (80 - y)² + y² = 6400 - 160y + 2y²
- Derivative: S'(y) = -160 + 4y
- Set S'(y) = 0: -160 + 4y = 0 → y = 40
- Sign chart: S'(39) < 0 and S'(41) > 0, confirming minimum at y = 40
- x = 80 - 40 = 40; Minimum S = 40² + 40² = 3200

5. Answer: L = 75 m, W = 75 m; Maximum area = 5625 m²

- Constraint: 2L + 2W = 300 → L + W = 150, so L = 150 - W
- Objective: A = LW = (150 - W)W = 150W - W²
- Derivative: A'(W) = 150 - 2W
- Set A'(W) = 0: 150 - 2W = 0 → W = 75

- Sign chart: A'(74) > 0 and A'(76) < 0, confirming maximum at W = 75
- L = 150 - 75 = 75; Maximum area = 75 × 75 = 5625 m²

6. Answer: x = 1; Minimum value = 2

- Objective function: f(x) = x + 1/x
- Derivative: f'(x) = 1 - 1/x²
- Set f'(x) = 0: 1 - 1/x² = 0 → x² = 1 → x = 1 (positive)
- Sign chart: f'(0.5) = 1 - 4 = -3 < 0 and f'(2) = 1 - 1/4 = 0.75 > 0, confirming minimum at x = 1
- Minimum value: f(1) = 1 + 1/1 = 2

7. Answer: x = 6 cm, h = 3 cm; Maximum volume = 108 cm³

- Constraint: x² + 4xh = 108, so h = (108 - x²) / (4x)
- Objective: V = x²h = x² · (108 - x²)/(4x) = (108x - x³)/4
- Derivative: V'(x) = (108 - 3x²)/4
- Set V'(x) = 0: 108 - 3x² = 0 → x² = 36 → x = 6
- Sign chart: V'(5) > 0 and V'(7) < 0, confirming maximum at x = 6
- h = (108 - 36)/(4·6) = 72/24 = 3; V = 36 × 3 = 108 cm³

8. Answer: r = 10 / (4 + π) ≈ 1.40 m; h = r/2 · π ≈ 1.40 m

- Constraint: 2h + 2r + πr = 10, so h = (10 - 2r - πr)/2 = 5 - r - (π/2)r
- Objective: A = 2rh + (πr²)/2; substitute h: A = 2r(5 - r - πr/2) + πr²/2
- Simplify: A = 10r - 2r² - πr² + πr²/2 = 10r - 2r² - (π/2)r²
- Derivative: A'(r) = 10 - 4r - πr
- Set A'(r) = 0: 10 = r(4 + π) → r = 10/(4 + π) ≈ 1.40 m
- h = 5 - r - (π/2)r = 5 - r(1 + π/2) ≈ 1.40 m

9. Answer: r = ³√(250/π) ≈ 4.30 cm, h ≈ 8.60 cm; Minimum surface area ≈ 348.7 cm²

- Constraint: πr²h = 500, so h = 500/(πr²)
- Objective: S = 2πr² + 2πrh = 2πr² + 2πr · 500/(πr²) = 2πr² + 1000/r
- Derivative: S'(r) = 4πr - 1000/r²
- Set S'(r) = 0: 4πr = 1000/r² → r³ = 1000/(4π) = 250/π → r = ³√(250/π) ≈ 4.30 cm
- Sign chart: S'(4) < 0 and S'(5) > 0, confirming minimum at r ≈ 4.30
- h = 500/(π · (250/π)^(2/3)) = 2r ≈ 8.60 cm

10. Answer: x = 150 units; Maximum profit = $9,250

- Profit: P(x) = R(x) - C(x) = (200x - 0.5x²) - (50x + 2000) = 150x - 0.5x² - 2000
- Derivative: P'(x) = 150 - x
- Set P'(x) = 0: 150 - x = 0 → x = 150
- Sign chart: P'(149) = 1 > 0 and P'(151) = -1 < 0, confirming maximum at x = 150
- Maximum profit: P(150) = 150(150) - 0.5(150²) - 2000 = 22500 - 11250 - 2000 = 9250
- The company should sell 150 units per week for a maximum weekly profit of $9,250